

Limits at the Edges

One-sided limits, holes and vertical asymptotes, and end behavior

Directions:

- Decide first *which question the notation is asking*: a value approached from one side, behavior at a zero of the denominator, or behavior as x runs off the axis.
- Acceptable answers are an exact number, $+\infty$, $-\infty$, or DNE. Write $+\infty$ or $-\infty$ when the values grow without bound in one definite direction — that is more informative than DNE.
- Show the rewriting step (factor, cancel, divide by the highest power, or split the absolute value) before you state the answer.

1. Let $f(x) = \frac{|x-3|}{x-3}$. Split the absolute value into its two cases and evaluate:

(a) $\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}$

2. The function $g(x) = \frac{x-4}{\sqrt{x-4}}$ is defined only for $x > 4$, so only one side of $x = 4$ exists. Evaluate $\lim_{x \rightarrow 4^+} g(x)$.

Answer: _____

3. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 2}{4x^2 + 7}$

Answer: _____

4. Let $h(x) = \frac{x+3}{x-5}$. The denominator is zero at $x = 5$ and the numerator is not, so $x = 5$ is a vertical asymptote. Track the sign of $x - 5$ on each side and evaluate:

(a) $\lim_{x \rightarrow 5^-} h(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 5^+} h(x) = \underline{\hspace{2cm}}$

5. A shipping cost in dollars for a parcel of weight x kilograms is modelled by

$$f(x) = \begin{cases} 3x - 1, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$$

(a) $\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}$

(c) Explain whether $\lim_{x \rightarrow 2} f(x)$ exists, and say what your two answers above have to do with it.

6. Evaluate $\lim_{x \rightarrow -\infty} \frac{2x^3 + x}{5 - x^3}$

Answer: _____

7. Let $k(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$. The denominator is zero at *both* $x = 2$ and $x = 3$, but only one of them is a vertical asymptote. Factor, cancel, and evaluate:

(a) $\lim_{x \rightarrow 2} k(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 3^+} k(x) = \underline{\hspace{2cm}}$

8. Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 - 3}{2x^3 + x}$

Answer: _____

9. Evaluate $\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{|x - 2|}$

(For $x < 2$ the quantity $x - 2$ is negative, so $|x - 2|$ is not $x - 2$.)

Answer: _____

10. **Challenge.** Let $r(x) = \frac{\sqrt{9x^2 + 4}}{3x - 1}$. Divide top and bottom by $|x|$ and keep track of the sign of x in each case:

(a) $\lim_{x \rightarrow \infty} r(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow -\infty} r(x) = \underline{\hspace{2cm}}$